Two-scale finite-size edge splitting at a repeated root of an extended Su–Schrieffer–Heeger chain

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Repeated decay roots in a finite topological chain produce power-law-prefactored edge profiles, but the collision of the usual root vectors makes a direct finite-chain projection singular. We resolve this collision for the open extended Su–Schrieffer–Heeger family with hoppings $u_0 = r^2$, $u_1 = 2r$, and $u_2 = 1$. A root-and-derivative boundary basis gives two positive projected energies in closed form. More importantly, a signed reversal maps the rescaled inverse hopping block to a rank-two matrix plus an explicit triangular tail. Singular-value perturbation then gives a uniform relative enclosure for every $r \in [1/5, 4/5]$ and integer $N \geq 61$. The two exact positive edge energies satisfy

$$E_{N,1} \sim \frac{(1-r^2)r^{N+1}}{N}, \qquad E_{N,2} \sim N(1-r^2)^3 r^{N-1},$$

uniformly on this interval. A min–max estimate places the third positive energy at or above the bulk-band distance $(1-r)^2$, proving that precisely four chiral in-gap levels occur in the certified range. Thus a single double bulk root produces two parametrically different finite-size hybridization scales rather than one common exponential splitting.

I. INTRODUCTION

The Su–Schrieffer–Heeger (SSH) chain was introduced as a model of polyacetylene [1] and is now a canonical setting in which chiral symmetry relates a bulk winding number to boundary-localized states [2]. Longer-range intersublattice hopping extends the winding number beyond unity and permits several boundary pairs. Such extended SSH (eSSH) structures have been studied dynamically [3] and realized in long-range-coupled photonic lattices [4]. For a finite device, the nominally zero-energy states at opposite ends hybridize. Their small energies, rather than the semi-infinite zero modes alone, determine whether the edge multiplet can be spectrally resolved.

Generalized Bloch theory provides exact bulk and boundary equations for finite-range free-fermion Hamiltonians [5, 6]. When a transfer or decay root is defective, its boundary profile acquires a power-law prefactor. This general mechanism is known and is not the subject of the present work. A recent analysis of the eSSH model derived exact semi-infinite edge states and accurate finite-chain expressions for two distinct complex-conjugate roots [2]. On the discriminant line where the two roots coincide, however, those two root vectors become linearly dependent. The finite-chain overlap matrix in the distinct-root representation is then singular, even though the physical two-dimensional edge space remains well defined through a confluent root-and-derivative basis.

Here we solve the finite-size splitting on one such repeated-root line. The model is simple enough that its off-diagonal block is $T_N = (rI + S_N)^2$, but the two smallest singular values occur at different polynomial orders in N. We first construct the exact four-state confluent Gram pencil. We then convert $r^{N+1}T_N^{-1}$, by orthogonal sign and reversal operations, into a symmetric rank-two matrix plus a triangular tail whose Frobenius norm is an explicit finite sum. This supplies a relative enclosure, not merely a residual estimate. General Toeplitz finite-section theory already gives qualitative splitting results [7]; the result below is the pair of coefficients, an explicit uniform remainder, and the exact in-gap count for this fixed Hermitian family.

II. OPEN REPEATED-ROOT CHAIN

Let $N \geq 3$ be the number of unit cells, with orthonormal orbitals $\{|A_j\rangle, |B_j\rangle\}_{j=0}^{N-1}$. We set the long hopping $u_2 = 1$ as the energy unit and take

$$u_0 = r^2, \qquad u_1 = 2r, \qquad r \in I := [1/5, 4/5]. \tag{1}$$

The open, single-particle Hamiltonian is

$$H_N = \sum_{j=0}^{N-1} r^2 |B_j\rangle\langle A_j| + \sum_{j=0}^{N-2} 2r |B_j\rangle\langle A_{j+1}| + \sum_{j=0}^{N-3} |B_j\rangle\langle A_{j+2}| + \text{H.c.} \tag{2}$$

There are no on-site or wrap-around terms. In the A-then-B basis,

$$H_N = \begin{pmatrix} 0 & T_N^T \\ T_N & 0 \end{pmatrix}, \qquad T_N = r^2 I + 2rS + S^2 = (rI + S)^2, \tag{3}$$

where $S_{jk} = \delta_{k,j+1}$. Hence H_N is real, Hermitian, and chiral. Writing the singular values of T_N as $0 < E_{N,1} \leq \cdots \leq E_{N,N}$, the spectrum of H_N consists of $\{\pm E_{N,a}\}_{a=1}^{N}$.

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For $z = e^{-ik}$, the periodic off-diagonal symbol is

$$Q_1(z) = r^2 + 2rz + z^2 = (z + r)^2. \tag{4}$$

The root $z = -r$ has multiplicity two and lies strictly inside the unit disk. With the convention of Ref. [2], the winding number is two. The distance from zero to either periodic bulk band is

$$\Delta_g(r) = \min_{|z|=1} |Q_1(z)| = (1 - r)^2 \geq \frac{1}{25}. \tag{5}$$

We use "in-gap" below to mean $|E| < \Delta_g$; the full interband gap is $2\Delta_g$.

III. COLLISION-SAFE FOUR-STATE PENCIL

Put $q = r^2$ and $z = -r$. On the left edge define the two real vectors

$$u_i = z^i, \qquad w_i = iz^{i-1}, \qquad i = 0, \ldots, N - 1, \tag{6}$$

with $w_0 = 0$. They are the root and derivative profiles at the double root. Let R reverse the cell order, $(Rx)_i = x_{N-1-i}$, and use Ru, Rw on the right edge. In the ordered basis (u, w, Ru, Rw), with the first pair on the A sublattice and the reflected pair on the B sublattice, introduce

$$A_N = \sum_{i=0}^{N-1} q^i, \qquad B_N = \sum_{i=0}^{N-1} iq^i, \quad D_N = \sum_{i=0}^{N-1} i^2 q^i,$$
$$\Delta_N = A_N D_N - B_N^2. \tag{7}$$

Direct summation gives the exact Gram and projected Hamiltonian matrices

$$G = \begin{pmatrix} G_L & 0 \\ 0 & G_L \end{pmatrix}, \quad G_L = \begin{pmatrix} A_N & -B_N/r \\ -B_N/r & D_N/r^2 \end{pmatrix}, \tag{8}$$

$$K = \begin{pmatrix} 0 & C \\ C & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -z^N \\ -z^N & -Nz^{N-1} \end{pmatrix}. \tag{9}$$

The cross block follows without an asymptotic truncation. Indeed, the double-root recurrence cancels all but the last two components:

$$T_N u = (0, \ldots, 0, -z^N, z^{N+1})^T,$$
$$T_N w = (0, \ldots, 0, -Nz^{N-1}, (N - 1)z^N)^T. \tag{10}$$

The finite sums in Eq. (7) can be evaluated without cancellation as sums, or in closed form as

$$A_N = \frac{1 - q^N}{1 - q},$$
$$B_N = \frac{q - Nq^N + (N - 1)q^{N+1}}{(1 - q)^2},$$
$$D_N = \frac{q(1 + q) - N^2 q^N}{(1 - q)^3} + \frac{(2N^2 - 2N - 1)q^{N+1} - (N - 1)^2 q^{N+2}}{(1 - q)^3}. \tag{11}$$

Define

$$m_N = NA_N - 2B_N, \qquad \phi_{1,N} = \frac{\sqrt{m_N^2 + 4\Delta_N} + m_N}{2}, \tag{12}$$

$$\phi_{2,N} = \frac{\sqrt{m_N^2 + 4\Delta_N} - m_N}{2} = \frac{\Delta_N}{\phi_{1,N}}. \tag{13}$$

For $N \geq 61$, $m_N > 0$ and $\phi_{1,N} \geq \phi_{2,N} > 0$. Expanding $\det(C - EG_L)$, or equivalently separating the two chiral parities of $K\boldsymbol{v} = EG\boldsymbol{v}$, gives the two positive Ritz energies

$$\boxed{\widehat{E}_{N,1} = \frac{r^{N+1}}{\phi_{1,N}}, \qquad \widehat{E}_{N,2} = \frac{r^{N+1}}{\phi_{2,N}}.} \tag{14}$$

Unlike a coincident-root substitution in a distinct-root basis, Eqs. (8)–(14) retain the nonzero overlap $-B_N/r$ and remain nonsingular at the collision.

IV. RANK-TWO INVERSE COMPARISON

The projected energies become controlled approximations to the exact singular values because the full inverse has a nearly rank-two form. Since $S^N = 0$,

$$T_N^{-1} = \sum_{d=0}^{N-1} (-1)^d (d + 1) r^{-d-2} S^d. \tag{15}$$

A column reversal and diagonal sign changes preserve singular values and map $r^{N+1} T_N^{-1}$ to the real symmetric matrix

$$(M_N)_{i\ell} = \begin{cases} (N - i - \ell) r^{i+\ell}, & i + \ell \leq N - 1, \\ 0, & i + \ell \geq N, \end{cases} \tag{16}$$

for $0 \leq i, \ell \leq N - 1$. With $a_i = r^i$ and $b_i = ir^i$, extend the nonzero branch of Eq. (16) over the full square:

$$F_N = Naa^T - ba^T - ab^T. \tag{17}$$

This matrix has rank two. On $\mathrm{span}\{a, b\}$ its nonzero eigenvalues have sum m_N and product $-\Delta_N$. They are therefore $\phi_{1,N}$ and $-\phi_{2,N}$, so the two nonzero singular values of F_N are exactly $\phi_{1,N}$ and $\phi_{2,N}$.

The remainder $R_N = M_N - F_N$ is supported on the opposite triangle:

$$(R_N)_{i\ell} = (i + \ell - N) r^{i+\ell} \quad \text{for } i + \ell \geq N + 1. \tag{18}$$

Consequently

$$\epsilon_N(r) := \|R_N\|_F = \left[\sum_{t=1}^{N-2} (N - 1 - t) t^2 r^{2N+2t} \right]^{1/2} \tag{19}$$
$$\leq r^{N+1} \sqrt{\frac{(N - 1)(1 + r^2)}{(1 - r^2)^3}}. \tag{20}$$

Let $\mu_{1,N} \geq \mu_{2,N}$ be the two largest singular values of M_N. Inverse singular-value ordering and Eq. (16) give

$$\mu_{a,N} = \frac{r^{N+1}}{E_{N,a}}, \qquad a = 1, 2. \tag{21}$$

Weyl's singular-value inequality applied to $M_N = F_N + R_N$ then yields

$$|\mu_{a,N} - \phi_{a,N}| \leq \epsilon_N(r), \qquad a = 1, 2. \tag{22}$$

V. UNIFORM ENCLOSURE AND ASYMPTOTICS

We now state the finite-size result. Every constant is explicit; the threshold $N = 61$ is sufficient and is not claimed to be optimal.

Theorem 1 *For every $r \in [1/5, 4/5]$ and integer $N \geq 61$, $\epsilon_N < \phi_{2,N} \leq \phi_{1,N}$ and, for $a = 1, 2$,*

$$|E_{N,a} - \widehat{E}_{N,a}| \leq R_a(N, r) := \widehat{E}_{N,a} \frac{\epsilon_N(r)}{\phi_{a,N} - \epsilon_N(r)}. \tag{23}$$

Moreover,

$$\frac{|E_{N,a} - \widehat{E}_{N,a}|}{E_{N,a}} \leq \frac{\epsilon_N(r)}{\phi_{a,N}} \leq \eta_N, \tag{24}$$

where

$$\eta_N = 150N \left(\frac{4}{5}\right)^{N+1} \sqrt{\frac{(N-1)(41/25)}{(9/25)^3}}. \tag{25}$$

The four closest-to-zero eigenvalues of H_N are exactly $-E_{N,2}, -E_{N,1}, E_{N,1}, E_{N,2}$ and lie strictly inside the bulk gap. Uniformly in r,

$$\begin{aligned} E_{N,1} &= \frac{(1-r^2)r^{N+1}}{N}[1 + o(1)], \\ E_{N,2} &= N(1-r^2)^3 r^{N-1}[1 + o(1)]. \end{aligned} \tag{26}$$

Proof. The Gram determinant has the Cauchy–Binet expansion

$$\Delta_N = \sum_{0 \leq i < j \leq N-1} (j-i)^2 q^{i+j} \geq q \geq \frac{1}{25}. \tag{27}$$

The geometric-series bounds

$$A_N \leq \frac{25}{9}, \qquad D_N \leq \frac{q(1+q)}{(1-q)^3} \leq \frac{16400}{729} \tag{28}$$

and $B_N \leq q/(1-q)^2 \leq 400/81$ show first that $m_N \geq N - 800/81 > 0$. They then imply, for $N \geq 61$,

$$\phi_{1,N} \leq m_N + \sqrt{\Delta_N} \leq 6N, \qquad \phi_{2,N} = \frac{\Delta_N}{\phi_{1,N}} \geq \frac{1}{150N}. \tag{29}$$

Combining Eqs. (20) and (29), maximizing the elementary factors at $r = 4/5$, gives $\epsilon_N/\phi_{2,N} \leq \eta_N$. Both comparisons

$$\eta_{61}^2 < \left(\frac{413}{1000}\right)^2, \qquad \left(\frac{\eta_{N+1}}{\eta_N}\right)^2 = \frac{16}{25}\frac{(N+1)^2}{N(N-1)} < 1 \tag{30}$$

are rational inequalities after squaring and cross multiplication. Thus $\epsilon_N < \phi_{2,N} \leq \phi_{1,N}$ throughout the stated domain. Equation (22) gives $\mu_{a,N} \geq \phi_{a,N} - \epsilon_N > 0$. Taking reciprocals in Eq. (21) proves Eq. (23); dividing by the exact energy instead gives Eq. (24). Because $\eta_N \to 0$ exponentially, the control is uniformly relative.

For the leading terms, the finite sums obey, uniformly on I,

$$\begin{aligned} A_N &= \frac{1}{1-q} + O(q^N), \\ B_N &= \frac{q}{(1-q)^2} + O(Nq^N), \\ \Delta_N &= \frac{q}{(1-q)^4} + O(N^2 q^N). \end{aligned} \tag{31}$$

Equations (12) and (13) therefore give

$$\begin{aligned} \phi_{1,N} &= \frac{N}{1-q}[1 + O(N^{-1})], \\ \phi_{2,N} &= \frac{q}{N(1-q)^3}[1 + O(N^{-1})]. \end{aligned} \tag{32}$$

Substitution into Eq. (14), followed by the relative enclosure, proves Eq. (26).

It remains to count the in-gap levels. Extend $x \in \mathbb{C}^N$ by zero to $\ell^2(\mathbb{Z})$ and let U be the bilateral shift. On the codimension-two subspace $x_0 = x_1 = 0$, no left-boundary output is lost, and hence

$$\|T_N x\| = \|(rI + U)^2 x\| \geq \min_{|\zeta|=1} |r + \zeta|^2 \|x\| = (1-r)^2 \|x\|. \tag{33}$$

The singular-value min–max principle implies $E_{N,3} \geq (1-r)^2$. Conversely, Eqs. (21), (29), and (24) give

$$E_{N,2} \leq \frac{150Nr^{N+1}}{1-\eta_N} \leq \frac{150(61)(4/5)^{62}}{1 - 413/1000} < \frac{1}{25} \leq (1-r)^2. \tag{34}$$

The middle strict inequality is again an exact rational comparison. The right-hand side is maximal at $N = 61$ because $N(4/5)^{N+1}$ decreases and $1 - \eta_N$ increases thereafter. Thus exactly two positive singular values lie below the bulk-band distance. Chiral symmetry supplies their two negative partners, completing the proof. □

The two splittings are parametrically separated:

$$\frac{E_{N,2}}{E_{N,1}} \sim \frac{N^2(1-r^2)^2}{r^2}. \tag{35}$$

Accordingly, the inner doublet $\pm E_{N,1}$ is much more weakly hybridized than the outer doublet $\pm E_{N,2}$ even though both descend from the same double decay root.

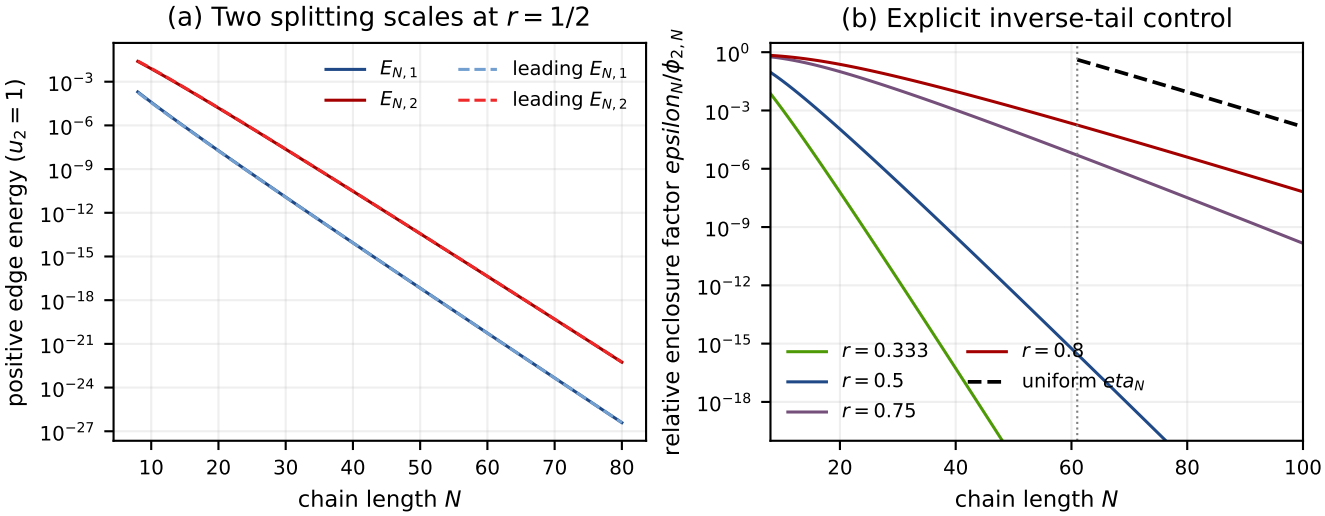


FIG. 1. Finite-size splitting and enclosure. (a) The exact positive edge energies at $r = 1/2$, evaluated stably from the two largest singular values of M_N, compared with the two leading laws in Eq. (26). (b) The computable worst-branch factor $\epsilon_N/\phi_{2,N}$ for four values of r. The dashed curve is the interval-uniform upper bound η_N, shown in its proven range $N \geq 61$; the vertical dotted line marks the sufficient threshold. The plotted points are direct evaluations of the displayed finite sums, with no fitted parameters. The plotting code was generated with the AI assistance disclosed in Sec. VI and checked against the tabulated data and analytic expressions.

VI. NUMERICAL CHECKS AND REPRODUCIBILITY

Figure 1(a) evaluates the exact small singular values via Eq. (21), avoiding loss of relative precision from a direct double-precision diagonalization of $T_N^T T_N$. Independent 110-decimal diagonalization of $T_N^T T_N$ was also performed at $r = 1/3, 1/2, 3/4$ and $N = 8, 12, 20, 40$. A separately assembled generalized eigenproblem from Eqs. (8) and (9) reproduced Eq. (14). Table I reports the larger relative Ritz error across the two positive branches. These checks illustrate convergence but are not used to infer the theorem.

Exact-fraction tests independently checked the Gram block, cross-Hamiltonian block, rescaled-inverse transformation, and tail norm in 40 cases obtained by crossing $N = 3, \ldots, 10$ with $r \in \{1/5, 1/3, 1/2, 3/4, 4/5\}$. The accompanying reproducibility package contains the high-precision validation program, the exact-fraction checker, the figure-generation program, and the tabulated figure data. The recorded environment uses Python 3.12.14, NumPy 2.3.5, SciPy 1.18.1, mpmath 1.3.0, and Matplotlib 3.11.2.

TABLE I. Maximum relative Ritz error $\max_{a=1,2} |E_{N,a} - \widehat{E}_{N,a}|/E_{N,a}$ from independent high-precision diagonalization.

r	$N = 8$	$N = 12$	$N = 20$	$N = 40$
1/3	9.73×10^{-5}	7.67×10^{-8}	1.35×10^{-14}	1.72×10^{-32}
1/2	1.16×10^{-2}	2.60×10^{-4}	3.20×10^{-8}	4.64×10^{-19}
3/4	2.97×10^{-1}	1.41×10^{-1}	1.33×10^{-2}	2.63×10^{-6}

AI-assisted research disclosure. OpenAI Codex with the `gpt-5.6-sol` model assisted with literature synthesis, analytic derivations, code generation and debugging, numerical cross-checks, and manuscript drafting. It was directed to preserve the fixed Hamiltonian, open-boundary convention, parameter quantifiers, and the distinction between exact statements and numerical checks. Outputs were challenged through independently constructed Hamiltonians and pencils, exact-fraction identities, high-precision diagonalization, and a separate reconstruction of the inverse-tail proof. The AI is not an author; this draft requires final review and approval by the named author before any submission.

VII. DISCUSSION AND LIMITATIONS

The repeated root does more than add a linear prefactor to a semi-infinite wave function. In a finite chain it creates two algebraically different end-to-end tunneling channels: one scales as $N^{-1}r^{N+1}$ and the other as Nr^{N-1}. The rank-two inverse extension exposes these channels directly, while its missing triangle gives an error that is exponentially smaller than both comparison singular values on the stated compact interval. The result therefore resolves both the coefficient and the polynomial prefactor that an $O(r^N \operatorname{poly} N)$ estimate leaves undetermined.

The interval $r \in [1/5, 4/5]$ keeps the repeated decay root uniformly normalizable and the bulk-band distance uniformly nonzero. The threshold $N = 61$ was chosen

for a simple interval-wide certificate rather than optimized; the numerical data show that the formulas are already useful at shorter lengths, but the same constants are not asserted there. We make no statement about disorder, interactions, on-site potentials, non-Hermitian hopping, experimental uncertainty, or a thermodynamic phase transition. General confluent Bloch profiles and qualitative Toeplitz splitting are prior results; the scope here is the explicit two-scale formula, uniform relative enclosure, and exact four-level count for Eq. (2).

DATA AVAILABILITY

No external data set is required. All numerical values are generated from the equations in this article. The scripts and tabulated data needed to reproduce Fig. 1, Table I, and the exact finite-sum checks are included in the accompanying reproducibility package.

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